

Supporting Youth Mental Health through Co-designing a Virtual Reality Experience: A Recipe-Based Co-design Model

Jin Ha Lee, Elin Björling, Ryan McLean, Anna Moreno, and Kung Jin Lee

ABSTRACT

There is a need for librarians to find ways to provide service to the youth population regarding mental health. Building on prior research about co-design and virtual reality in public libraries, the research team aimed to create a positive teen mental health experience by co-designing a virtual reality space through the library system. Eight librarians/library staff representing three participating sites were interviewed at the beginning and the end of the project. We discuss various opportunities and difficulties that emerged in this project to better understand the role and perspective of librarians and then examine implications for those who are considering implementing similar youth programs at their institutions. On the basis of the findings, we propose a recipe-based co-design model to better support librarians by allowing flexibility while clarifying which core elements need to be preserved in the co-design process.

The COVID-19 pandemic brought significant changes to people's lives, including the youth population, who had to adjust to new social distancing restrictions and online/hybrid schooling. They lost opportunities to interact with their friends in person and had to swiftly adapt to new modes of learning, all while dealing with uncertainties and fear, which had a profound impact on their mental health. These effects were especially impactful to those who were already vulnerable (Courtney et al. 2020; Power et al. 2020). The COVID-19 pandemic has severely exacerbated the negative effects of the lack of mental health support among teens with a steep increase in anxiety and depression among adolescents (Racine et al. 2021). The pandemic also impacted the kinds of library services safely offerable to the general public, including youth. Many libraries had to limit their services, switch to programs that could be offered in an online mode, or close altogether (Cleave and Geijsman 2020; Goddard 2020; Kretz, Payne, and Reijerkerk 2021). There is a pressing need for libraries to find ways to continue providing service to the youth population in this challenging time. In

This work was made possible in part by the Institute of Museum and Library Services, award CAGML-247285-OMLS-20.

The Library Quarterly: Information, Community, Policy, volume 94, number 2, April 2024.

© 2024 The University of Chicago. All rights reserved. Published by The University of Chicago Press. <https://doi.org/10.1086/729230>

particular, there seems to be an opportunity for public libraries to serve as accessible hubs to support the mental health of youth patrons, as they are well-embedded in various communities and are considered trusted institutions for finding reliable information (Geiger 2017).

The Seattle Public Library, in collaboration with the University of Washington, launched an initiative supported by the Institute of Museum and Library Services (IMLS) to integrate co-design and virtual reality (VR) technology with the goal of engaging teens, aged 14 to 18, in the design and development of a VR experience focused on supporting and improving their mental health. The number of youth library programs involving co-design has been increasing, as these programs have shown positive impacts on youth by making them feel valued and listened to and offering the opportunity to develop teamwork and leadership skills and contribute to their community (Druin et al. 2009; Yip and Lee 2018). These co-design programs focused on a wide range of digital media, including 3D printing, Arduino, e-textiles, and location-based games. VR has also been increasingly adopted in libraries for informal learning and social engagement (Lee et al. 2020). VR, in particular, has been shown as an engaging technology for adolescents with the potential to support mental health by reducing stress, promoting mindfulness, and regulating emotions (Flores et al. 2018; Björling et al. 2019). RESeT, a virtual reality environment designed with teens through the Seattle Public Library to help with stress (Björling 2019), and VR programs in Washington State Libraries (Dahya et al. 2021; Lee et al. 2020) are two examples of co-design programs at public libraries incorporating VR. Building on prior research on co-design and VR at public libraries, the research team aimed to create an opportunity for improving youth's mental health during the prolonged global pandemic.

The co-design program was implemented in three different public libraries. Two of the three were urban public libraries in Seattle and Washington, DC, with the third being a rural public library in La Grange, Texas. This paper examines our understanding of the perspective and role of librarians and library staff involved in this project, while the detailed analysis of the design process and outcome will be shared elsewhere. We interviewed librarian and library staff participants representing all three sites before and after the VR program.

RQ1: What do librarians and library staff perceive as the opportunities and challenges of offering youth co-design programs to support mental health?

RQ2: What tensions emerged during the co-design program, and what are the implications for librarians and library staff considering implementing such programs?

Relevant Literature

Mental Health, Teens, and Libraries

We are facing a teen mental health crisis in the United States. Nearly 10% of adolescents in the United States, estimated at 2.3 million, suffer from depression, with minority and poor youth

disproportionately affected (Giota and Gustafsson 2017; National Institute of Mental Health [NIMH] 2019; Ivey-Stephenson et al. 2020). Exacerbating this crisis is the fact that mental health care demands far exceed capacity, leaving a large proportion of adolescents without access to mental health services (NIMH 2019; Golberstein et al. 2020; Ivey-Stephenson et al. 2020). The need for community-based support for adolescent mental health has never been more dire, as the effects of the COVID-19 pandemic have exacerbated an already dire mental health situation. Globally, clinical rates of anxiety and depression have doubled in adolescents during the COVID-19 pandemic (Racine et al. 2021), which prompted the declaration of a national emergency (American Academy of Pediatrics 2021). As a consequence of pandemic-related school closures, social isolation, and loneliness, adolescent mental health issues, including rates of depression (Wang et al. 2021), have surged.

As an accessible, community-based setting, libraries are uniquely positioned as potential hubs to access mental health support. Mental health is already emerging as a topic of interest in many communities: In a recent survey of 50 different library institutions in the United Kingdom, 75% said they had taken on some activity to support mental health and well-being during the 2020 COVID-19 pandemic (Cox and Brewster 2020). Seventy-nine percent of respondents said that their librarians were offered short courses run by their university on mental health and well-being. Given the scale of the mental health concerns in adolescents, it is likely impossible for librarians serving teens to avoid the topic of mental health, yet only 17% of libraries indicated that staff were formally trained on mental health. Sixty-eight percent of libraries said that building a sense of belonging was a main issue they tried to address during the pandemic, and 20% said they were actively working to reduce stigma around mental health. Taken together, the findings suggest that librarians are interested in providing mental health support for the patrons but lack sufficient training and administrative support to do so confidently.

Participatory Design and Libraries

Participatory design (PD) is a user-centered design method (Sanders 2002) that has been adopted and utilized in libraries to develop various programs and digital learning experiences (Yip, Lee, and Lee 2020). Prior examples include various stakeholders such as students, faculty, and librarians coming together to design physical or virtual learning commons (Somerville and Collins 2008; Santos et al. 2017), digital learning activities (Yip et al. 2020), and organizational structures and communication systems (Somerville and Howard 2010) and develop a public library (Dalsgaard and Eriksson 2013).

Collaboration between youth and librarians often occurs in the form of advisory boards where the youth advise librarians on the design of programs. Linda Camino (2005) points out several challenges with this approach, namely the potential over-dependence on youth to run the programs, adults not yielding power to youth, and bias linked to stereotypes about

youth. Camino discusses three key elements for successful youth and adult partnership, which are fundamental in participatory librarianship with youth: 1) learning communities (Senge 1991) as a group of people who help each other build new skills and capabilities; 2) communities of inquiry where people redefine their roles to be both creators and users (Friedman 2006); and 3) communities engaging in reflection in practice (Schön 1987). Jason C. Yip, Kung Jin Lee, and Jin Ha Lee (2020) also discuss how the librarian's role shifts between "supporter of youth" and "designer with youth" with regard to four different aspects of design: relationship building, facilitation, design-by-doing, and idea elaboration. They argue that frequent transitions occur between these roles during the co-design process and emphasize the importance of providing pre-service librarians opportunities to participate in a realistic participatory design process.

Additionally, with the COVID-19 pandemic and subsequent shift to online programming in many libraries, Kung Jin Lee et al. (2021) present a conceptual model of conducting synchronous participatory design with youth, building upon theories of improvisation. They suggest that key features of improvisation—including reflexivity, transgression, tension, listening, and interdependence—play a critical role in online synchronous co-design, which poses additional uncertainties and challenges compared to in-person co-design. Two of the three library sites in our study also had to engage in online synchronous co-design, and thus, a prominent need for improvisation emerged (as further discussed).

Teens, Libraries, and VR

There has been a long interest in understanding teens' perceptions of libraries. By using both quantitative and qualitative methods, Vivian Howard (2011) shared findings about teens' perceptions of libraries. While the survey results showed overall satisfaction, the focus group analysis showed critical dissatisfaction about barriers to relationships between library staff and teens. Studies have called out the importance of responding to the teens' needs to foster and maintain habits of library use, indicating a need for better technology access (Agosto et al. 2016) and more opportunities to connect with tools that can leverage their learning in the library (Braun et al. 2018). Among the different tools, VR has been a popular modality for teens due to the social engagement opportunities and broad immersive activities within VR (Maloney, Freeman, and Robb 2021). As interest in VR continues to grow through social media, gaming, and entertainment industries, more public libraries are offering an opportunity for their patrons to experience VR through VR-specific programming (Lee et al. 2020; Dahya et al. 2021). Public libraries serve as a community hub in a wide variety of communities, including rural and urban areas and thus are an excellent venue for disseminating new technologies, such as VR, to the general public. VR is also being adopted in more academic libraries (Pope 2018; Frost et al. 2020) and specialized libraries, such as health sciences libraries (Lessick and Kraft 2017), for its unique learning opportunities and user engagement.

Despite the opportunities that VR provides, many challenges still exist. Some are related to technology; for instance, Negin Dahya et al. (2021) point out that inequalities embedded in the technology itself, as well as perceptions of the technology, resulted in unequal access for marginalized groups such as women or immigrants in their study. Kung Jin Lee et al. (2020) also discuss issues related to the experience of VR in a public place like the library, where not everyone feels safe or comfortable being on public display. Practical challenges, such as how to deal with multiple people sharing the VR headset, scheduling the use of VR, setting limited time for each patron, and providing support for the play session to ensure safety for the patron, were also mentioned as hurdles in a public setting. For the future direction of the VR programs in libraries, Lee et al. (2020) discuss the importance of library patrons participating in the design of VR content beyond accessing VR as consumers; our project attempts to include youth patrons in libraries to co-create VR content without requiring extensive technical knowledge or coding background.

Study Design and Method

Motivation

This research study was part of a larger project to explore the co-design of a VR environment intended to support teen mental health. Three library sites (Seattle Public Library, DC Public Library, and Fayette Public Library, Museum and Archives) engaged teens in learning about VR technology and designing their own VR environments through their public library system. We wanted to understand librarians' experiences being part of this program and how their perspectives changed as a result. To answer this question, we employed qualitative pre- and post-program interviews.

Program

The focus of the program was to create a positive teen mental health experience by co-designing a VR space through the library system. The goal of the librarians was to create close relationships with teens and community stakeholders by centering the discussion on mental health and providing a digital learning experience. The three pillars that guide the projects are: teen mental health, co-design, and VR.

At each of the three library sites, librarians (including library staff) partnered with a group of 10 to 12 teens and engaged in a series of co-design sessions reflecting on mental health and generating ideas on how the VR program could support their mental health and well-being. All the teens were between 14 and 18 and had a range of experiences with VR, with some completely new to VR while others had played VR games with friends. They partnered with undergraduate student interns who worked as the VR developers. These interns took ideas generated from the co-design sessions to create successive iterations of the actual VR experience. These iterations were demoed for the teens, who provided feedback for subsequent changes.

There were two interns assigned to each site from the University of Washington and Howard University.

Librarian Sample

This study included eight librarians (including library staff) across the three sites. The eight librarians were interviewed at the beginning and the end of the project for a total of 16 interviews. All the librarians had experience working with teen patrons, and two also had experience with co-design and VR.

Individual Interviews

We conducted semi-structured interviews with the librarians via Zoom prior to beginning and after the completion of the program. The pre-program interviews focused on the librarians' expectations, opinions, perceived values, and comfort level with implementing a new VR co-design program for supporting mental health with youth. We also elicited concerns about the program. The post-program interviews invited librarians to reflect on their initial concerns and to share any insights that arose during or after the conclusion of the program. The interview protocol is included in the appendix. Questions included: Given your original concerns and impressions about the project, what has surprised you based on your experience? Given your experience, has your level of comfort changed regarding running library programs to support patrons' mental health? Interviews were recorded and transcribed for data analysis.

Community of Practice Discussion

Another data source for this analysis was a "Community of Practice" discussion with project staff and librarians. In this discussion, the project team was asked to use a shared MIRO board (fig. 1), a platform for online collaboration, to reflect on what went well and what they would change about the program. Example questions include: What are some lessons learned through your experience implementing this program? What are the biggest likes you experienced while implementing this program? What are some design ideas you have for making the program more engaging, impactful, and/or easier to implement?

Qualitative Analysis

A modified, collaborative thematic analysis (Clarke et al. 2015) was conducted in which four of the five authors collaboratively analyzed the data. The authors conducted all four phases of a reflexive, thematic analysis including: familiarization, coding, theme generation, and theme review. During familiarization, authors were guided by the research question "What was the experience of library staff during this program?" and assigned individual interviews and associated transcripts to explore for salient excerpts. Initial codes were then applied to these excerpts to help describe participant experiences. All excerpts and codes were captured in a



Practice discussion

shared MIKO board to facilitate a collaborative analysis. These excerpts and associated codes were then collaboratively reviewed, discussed, and organized by all the authors until core themes emerged and were formalized (theme generation). Example themes include “apprehension” and “the library as a community.” Upon writing the current paper, these data were organized into more practical, categorical sections/themes in an effort to summarize successes and challenges, utilizing direct excerpts from several of the themes.

Findings

Opportunities and Challenges

In this section, we answer the first research question on librarians' perceptions of opportunities and challenges in offering youth co-design programs to support mental health. At the start of the project, many librarians expressed potential challenges related to this project, including facilitating programs with a mental health aspect, conducting co-design with teens, and the

novelty of VR technology. In addition, the onset of the COVID-19 pandemic occurred after libraries had committed to the program but before the actual program began, adding another layer of challenge and uncertainty. Fortunately, each of these challenging elements provided learning opportunities for project team members, engagement among the teen participants, and an overall sense of value for the librarians involved. Below, we illustrate each challenge area as well as how those challenges were overcome.

Mental Health Support Programs at Libraries

Librarians' attitudes notably shifted when comparing pre- to post- interviews. When asked about their roles, experiences, and concerns in "supporting patron mental health," many pre-interviews described prior or anticipated future situations where patrons were suffering from poor mental health. These interviews revealed a high desire to support patrons but low confidence in their ability to effectively help patrons in these circumstances. After learning about mental health during the course of the program, many librarians expressed a shift in perception to seeing mental health issues as something that all people experience and believing that an approach supportive of mental health and well-being could be adopted by many library services.

Overall, the greatest concern librarians expressed was engaging with the topic of mental health. This was true even for librarians with prior experience in co-design, VR, and teen programming. The majority of facilitators voiced a similar concern that "saying something wrong" during the course of co-design exercises might inadvertently harm the teen participants, although both the "saying something wrong" and harm were ambiguous. A commonly cited contributing factor to this concern was a lack of prior training or education specific to mental health, which may have created a general insecurity about engaging in this topic. Librarians P4 and P2 shared, "[I am] afraid of, sort of like upsetting the people who are participating and doing more harm than good . . . we have a social worker for a reason" (P4, Pre-Interview); "you don't want to be in that paralyzing position and you want to be able to help youth correctly" (P2, Pre-Interview).

However, Librarian P8 discussed how broadening the librarian's perspectives on mental health to include wellness could potentially help them feel more comfortable about participating in the program:

I think it really depends on the program itself, I know that I am not like a medical professional so I can't give specific advice as to this is how you need to be dealing with this issue that you have, but if it's about wellness, if it's about how do you, you know find peace, how do you find things that relax you, that I'm perfectly comfortable with . . . if it's something to bring awareness, or to bring familiarity on the topic . . . getting the knowledge necessary to identify any concerns you may have. (P8, Pre-Interview)

At the beginning of the project, the research team gave a presentation about mental health and co-design to support the librarians. In the post-project interviews, multiple librarians shared how this presentation helped broaden their overall perspectives on mental health, helped “destigmatize/normalize mental health,” provided tools to “address mental health without being too ‘heavy,’” and provided useful vocabulary for discussing the topic, according to notes from the MIRO board. Librarian P8 stated, “This idea of mental health being on a sort of—not being binary—is not you either have it or you don’t, but being on this scale of you know, maybe sometimes it’s a little lower, sometimes it’s a little higher, so just that framework really helped guide a lot of the decisions” (P8, Post-Interview).

The concerns around addressing mental health also eased once the program and co-design sessions began. Reflecting on this, several librarians explained that while they had originally associated “mental health support” with crisis situations such as teen self-harm and suicide, during the sessions they were able to shift their perspective toward everyday mental health and well-being: “Going through this project, it really made me think, it’s easier to talk about mental health than I thought, because we do it all the time, naturally, and it was kind of cool to get that perspective of thinking about mental health as a spectrum, which makes total sense” (P2, Post-Interview).

Librarian P7 talked about a turning point when they realized that, rather than creating an intervention, the goal of the program was to create a space where youth and librarians could talk about mental health. (“Teens want to talk about and design around mental health,” from the MIRO board.)

Multiple librarians felt that libraries had an important role in their communities and expressed the importance of listening to and understanding patron needs and using this knowledge to tailor the library’s programming, including the types of services and technology access the library prioritizes. The need to address mental health, especially for youth patrons, was seen as inevitable and important, as P6 shares, “[In my] experience of just working with teens, it kind of inevitably crosses over into mental health, just because it’s such an emotionally turbulent time and without a doubt you’re going to have some teen coming in and maybe they’re being bullied or something like that, they’re just going to talk about it at some point” (P6, Pre-Interview). Facilitating the co-design sessions gave librarians a first-hand confirmation of the potential demand, at least among teens, for programs supporting mental health. Further, librarians acknowledged the potential role librarians and libraries could play in democratizing mental health care. Librarians stated,

Yeah, I mean I think it’s super important to be able to, you know, maybe democratize some of these skills and some of these frameworks and information. Because again, like you think about the topic mental health and you automatically assume that someone in the room needs to be an expert to talk about it, or to be able to facilitate a discussion,

but that's not the case at all. As soon as we started to just open up spaces, especially in places like libraries, it really helps. (P8, Post-Interview)

Hypothesis: when given the space, teens will create things that are good for their mental health. (MIRO board)

Librarian P1 also had the realization while working with teens that librarians could potentially create more space for mental health for their patrons and provide the time and space to talk about difficult topics.

I don't think that we fully grasp how they are feeling and how they reflect on that. We have not provided them enough time and space to communicate those things, and by not doing that, they simply don't think about it. [. . .] these [mental health] are things that you have to be proactive about because we're so used to be trained to be reactive. We're trained to, okay, if this happens call this person, do this, or refer them to somebody, or God forbid call their parent, [. . .] we need to create that safe space where that's something that we can talk about all the time. (P1, Post-Interview)

P3 also reflected on the relationship between public libraries and social services and how library staff's work inherently becomes tied to mental health:

The way public libraries actually function, because they are one of the last community spaces left . . . it's very much part of the fabric of social services . . . library staff end up being [at] one of the places where people with no training on how to deal with people being on the edge or experiencing problems that affect social behavior and even personal behavior . . . we really have to sort of deal with it and navigate that world . . . It's been on my mind a lot, how much of what we do really is about everyone's mental health and how that is also tied to a lot of other factors. (P3, Post-Interview)

Co-design of VR Content

Many librarians expressed apprehension about their ability to conduct co-design sessions and/or learning to use the actual features and functionalities of the devices. Many expressed a desire for highly detailed, structured program support information. Some libraries did encounter challenges with the technology, but these tended to be technology adjacent to the VR devices (laptops, internet connections, etc.) or mundane aspects of the technology (cables, batteries) relative to what was expected. Attitudes toward co-design were more divergent after the program. When a sense of shared creation could be attained, co-design was seen as highly valuable and enabling, both in engaging teens and in supporting mental health. Co-design was perceived

as the key enabler, and more so than VR. Some libraries experienced technical challenges and were inhibited from reaching a sense of shared creation by these challenges.

Most of the librarians in this cohort said that their formal graduate program education did not include design methods coursework. While two mentioned prior experience with co-design, the remaining six could not recall any such significant experience. When asked directly if they “saw” or considered themselves to be designers, all responded no, including those who mentioned prior work that could reasonably be considered co-design, such as creating services alongside patrons. One of the librarians (P2, Pre-Interview) described design/co-design as being “very specific, for a specific group,” and another struggled to identify with the term “designer.” Librarian P3 explained, “I always see myself as a passive observer of the design process. Sometimes I feel like I’m a little too analytical to get my hands dirty and jump in. I have done it, but [designer] is not a label I attach to myself” (P3, Pre-Interview).

Despite a lack of co-design experience or design identity, the majority of librarians anticipated that engaging in co-design would create positive outcomes for the teen teams. Librarians anticipated teens increasing their ability to collaborate and work in teams through autonomy and ownership of their ideas brought by co-design. Librarian P4 shared, “Co-design helps develop a number of different skill sets that are important for a teen as they enter into adulthood, so you know there’s critical thinking, creative thinking” (P4, Pre-Interview).

The primary co-design challenge was the anticipation of a scenario where the lesson plans lacked enough structure and details to enable the librarians to facilitate co-design effectively. Librarians also expressed a discomfort in the general uncertainty and lack of a predetermined outcome inherent in co-design. Librarians expressed a desire for more structure and planning, which they believed would help with the success of the program. Librarian P5 shared their expectation for co-design: “The structure has to be very carefully presented for it to be effective. [. . .] [It] really has to be thought out and all of the ways in which people connect or don’t connect have to be certain. [. . .] And then there’s some accountability, of course. [. . .] Without accountability, without real engagement from all parties, you’re not really at co-design” (P5, Pre-Interview).

However, during the co-design experience, several adjustments were made to ensure that everyone was able to participate effectively and comfortably. For instance, at the Seattle Public Library site, initially the design activities consisted of VR sessions that ranged from 45 minutes to an hour. Feedback collected after earlier sessions revealed that some participants were experiencing VR-induced motion sickness, especially when using apps with lateral movement. Responding to this feedback, the design team adjusted future sessions to reduce the amount of continuous time participants spent within VR, opting instead to expand the use of digital tools like MIRO as well as “offline” tools like sketching on paper. Other accommodations included swapping VR apps that the group found difficult to use (e.g., Mozilla Hubs) with alternatives (e.g., CoSpaces). We believe this responsiveness to early suboptimal experiences helped

sustain engagement from the teens at a level that librarians noted as higher than teen programs they had facilitated previously. P3 additionally shared how having a co-design participant who experienced the program at another site helped with the process:

We had a really hard time until [the student researcher] came and helped us out. While we had watched some of the recordings [but] none of us had participated in something like this before . . . I always feel in co-design you need at least one person to be exactly on the guide-post and where you want to be hitting to make sure you're always moving in that direction even if how you get there is different. You need to know where you're supposed to be at the end . . . The technology wasn't super difficult; we were able to get that down. It was the co-design piece for us [that was difficult]. (P3, Post-Interview)

P4 also noted that once they were given the structure to facilitate the co-design process, they initially felt the lack of flexibility in the process ("The templates made it so much easier but needs a bit more ambiguity to mold to each library's needs" [MIRO board]) but were able to successfully run the sessions after clarifying the co-design goal and adaptable elements: "[For] each session, okay, this is what we're hoping to accomplish in this session and we're like, okay, so . . . as long as we accomplish this, we're good" (P4, Post-Interview).

In interviews conducted at the conclusion of the project, the majority of librarians described their experiences positively. P3 was content that the co-design practices were getting integrated into their normal programming activity, and they were using it more in the organization. P8 shared that they were surprised by how positive the co-design experience was, despite their initial concern about doing co-design via Zoom:

I was pleasantly surprised that we were able to find tools like MIRO that allowed us to collaborate, even if it is in this very different environment, you know. I wasn't able to see a lot of the teen's faces as they were somewhat shy at the beginning, and so I was surprised [by] the level of engagement we were able to sustain for as long as we did, through these co-design activities. I think that was really a cornerstone of making every meeting really fun and that they were able to derive value from each one of our meetings. (P8, Post-Interview)

Some of the challenges that emerged in co-design related to the nature of current VR content development and tools. For this cohort, groups were partnered with undergraduate VR developer interns who helped realize the teens' design ideas in the VR space through an iterative process. This process allowed teens to participate in VR content creation even without extensive coding experience. While this back-and-forth slowed the design process compared to if the teens had been directly creating their final deliverables, the librarians felt

that working with VR provided a unique co-design opportunity not just for teens but also for undergraduate interns who worked as professional developers: P8 noted that “it’s completely immersive so you can take a teen out of whatever situation they’re in and for a brief second put them in any sort of environment . . . have them be in a relaxation space or have them be in a learning space or just a fun space so it’s just that embodied experience that is so transformative in your learning” (P8, Pre-Interview). The MIRO board also showed comments regarding biggest likes such as: “seeing the excitement of the teens about their final product”; “giving undergraduate students opportunities to have a real world [development] experience.”

One librarian, P1, suggested that co-design participants postulated that the co-design itself, which requires participants to open up, share ideas, and work together toward a goal, was supporting teen mental health: “And we were talking about mental health and co-design and VR and everything like that. One of the teams even said, well, I think, designing together was beneficial for my mental health” (P1, Post-Interview).

Value of VR in Library Programming

Considering VR in the larger scope of library programming, prior to this program, many librarians had been looking forward to exploring how VR and its capabilities might benefit patrons. Several framed the relationship between libraries, communities, and novel technology like VR as symbiotic, expected, and common. Concerns around VR were more systemic in nature, such as difficulties common to bringing any new technology into the library, or inequality in access to technology within their communities. Librarians were somewhat undecided on whether they imagined themselves incorporating VR into future programming. While the overall sentiment was positive, the enthusiasm for VR seemed to be overshadowed by enthusiasm toward co-design and incorporating supportive approaches toward mental health into library services and programming.

As used in this program, VR was described as both “engaging” as well as “a potential barrier across settings.” Most librarians agreed that VR development supported the library agenda of technology literacy for both patrons and librarians. However, many also felt that VR technology was overwhelming at times, even with substantial technical support. The following quote from the MIRO board suggests substantial time is needed for preparing to use, rather than learning how to use, VR: “The amount of time it took to plan sessions and prepare the technology for each session (was longer than expected); the amount of time it took for the teens and staff to learn the technology (was shorter than expected)” (MIRO board).

Some librarians found that VR was powerful in drawing teens into the library and co-design program. P4 and P5 shared how much they enjoyed the VR experience with the teens, especially during the pandemic, when physical gatherings were restricted and online interactions in apps like Zoom felt limited:

We were in VR chat, like that was probably the most fun that we had, like that was the most natural interaction we had, and it was really cool, like you know, playing with them in that space. (P4, Post-Interview)

To use VR was a really interesting way to co-design, specifically in that, it was like manipulating what feels like a real space, so it's almost like designing in the community or designing physical space which is something I think a lot of people don't think about, but they have a really fun time with. (P5, Post-Interview)

Biggest likes: combining everyone's thoughts into one VR experience. (MIRO board)

However, one librarian suggested that VR was not the reason teens were engaged in the program. P1 suggested that although VR increased tech literacy within their library, it was not the drawing point for participation, and mentioned that they felt it was excluding some teens. While this librarian could recognize that their comfort with VR technology had improved, the added value of it for teen programming was less apparent: "So as far as VR, yes, I feel a lot more comfortable with it, since I had no experience to begin with, but I don't know that we would use VR again as the method [to draw in teens]" (P1, Post-Interview). The ideas from the MIRO board on how to make the program more engaging (e.g., "More implementation of mental health ideas and mapping them to VR interactions") suggest that perhaps some librarians experienced difficulties understanding how the VR content being provided had connections to mental health. They felt the need to explore more content with the teens and make connections of how the interaction within VR would influence the teen's mental health ideas ("more exploration in VR—have teens involved in picking games/experiences to play together"). Further, they pointed out the varying degree of comfort and expertise with VR technology among the teens and suggested "leveled activities for teens and groups that are more comfortable with the tech and VR mechanics."

Additionally, a community member who worked with ethnically diverse youth suggested to P1 that a VR program was not going to meet the interests of his community of teens. P1 added that there could be other programs the youth would be interested in that could still teach skills: "It's just that this [VR] was not what was going to draw them in. And it wasn't what they felt they needed." This sentiment aligned with what Dahya et al. (2021) discovered in their VR project with Washington State Libraries; they argued for a more active and intentional recruiting approach to include users from marginalized groups, because patrons often tend to view VR as a gaming tool that is targeted for "techie" people and not for them. They suggest connecting VR with larger library events, making it part of a themed activity (e.g., art, science), and designing interactive wraparound programming with other related activities that could help make it more enticing to wider audiences. VR also introduced some privacy challenges. Librarians encountered some parents/caregivers who were hesitant to let the youth

participate because they had to sign up and obtain a Facebook account to be able to use the Oculus Quest VR headset.

Finally, librarians noted the potential complexity of incorporating professional VR developers into the design process. For example, the external developer may have detracted from a sense of ownership in the project design (e.g., misinterpreting some of the teens' ideas) or caused a delay in the VR development cycle, which may have detracted from the natural flow of collaborative co-design.

Overall, the VR aspect garnered mixed reactions from the librarians. Some felt very enthusiastic about its appeal to teens and expressed desires to continue incorporating it in their future services, while others felt uncertain as to whether they will continue to include it in teen programming.

Implications for Future Programs

In this section, we answer the second research question on tensions that emerged during the co-design programs and the implications for librarians considering implementing similar programs, proposing a recipe-based co-design model.

Role of Librarian and Libraries in Providing Mental Health Support

Establishing libraries as places to provide mental health support presents a huge opportunity to improve the mental health of teens. Teens rarely have easy access to mental health resources, especially teens who live in rural areas or are part of marginalized communities (US Department of Health and Human Services 2001). Even where there is greater access, stigma against seeking mental health support in communities or cultures could still prevent people from seeking help (Bulanda et al. 2014). Because of the way libraries are embedded into many communities, it is common for librarians and teens to regularly see each other, leading to a sustained relationship (Subramaniam 2016), as with a teacher, coach, or another community mentor. Library programming allowing teens to discuss mental health with librarians might help normalize mental health support as a regular, everyday matter. Rather than just addressing crisis situations, mental health support might be considered similar to health activities "for all," such as daily walks, eating well, and getting quality sleep. Despite initial concerns and hesitancy, the librarians who participated in our program were unanimous in their support of continuing to offer similar programs focused on mental health. Moreover, librarians also speculated that facilitating such programming might help improve their own mental health as well.

Value of Co-design Process

The co-design process provided a context for teens and adults to engage in conversations around mental health openly and comfortably. While the design outcomes were important

contributors to participants' sense of accomplishment, the design process itself was also very important. Librarians built rapport and connections through the co-design process and learned that others also experience challenges and employ coping strategies similar to their own. Learning about what fellow participants found comfort in and liked throughout the co-design process allowed participants to incorporate those elements in the design outcomes for others and establish connections. Librarians shared how being vulnerable by sharing their own personal stories and feelings about mental health seemed to encourage teens to do the same. Most librarians also noted how successful the program was in terms of the continued high engagement from teen participants. The interview data suggests that an important factor for that success was the connections participants built by engaging in design activities together.

It is important to note that the sense of design ownership, a key value of the co-design process, may have been unintentionally compromised for teens because of challenges related to VR. Although VR provided an engaging and novel platform to work with for some sites, for other sites, adopting this technology may have excluded some teens. Given the desire for inclusivity and diverse teen engagement within public libraries, it is important to consider the possibility of excluding teens uncomfortable with or uninterested in VR.

A Recipe-Based Co-design Model

One of the biggest challenges librarians expressed relative to co-design was finding the right amount of structure to provide confidence to librarians with little co-design experience, while leaving enough flexibility and freedom for all librarians to adapt the program to their specific contexts. The co-design process at the Seattle Public Library site was considered very successful both in terms of youth engagement and the outcome. Based on interviews and observation of the co-design sessions, a major contributing factor may have been that the librarians at this site had prior co-design experience. As difficulties arose, these librarians were able to adapt and "improvise" (Lee et al. 2021) as needed without hesitation, changing aspects of co-design on the fly in response to contextual factors such as technology, setting, and participant feedback. In initial interviews and conversations that occurred as part of the program, librarians from other sites expressed a desire for detailed structure and plans, which they felt important to help them get started. Later, some of this structure came to be perceived as restrictive rather than enabling—especially where techniques or tools did not seem to fit local circumstances. For librarians less familiar with co-design, it seemed unclear both "how" and "how much" they needed to either adhere to or adjust the provided plan.

Based on this finding, we propose the analogy of a "recipe" for future co-design curriculum, which we hope will support further adoption of co-design among libraries with varied levels of co-design experience. In cooking, recipes describe how to reproduce certain dishes and are often written with the assumption that individuals have no prior experience making that dish.

Even lacking in experience, a cook may be able to substitute ingredients called for in the recipe while still producing an outcome true to the original nature of the dish, so long as those substitutions are thoughtfully made. Meanwhile, other seemingly minor substitutions may cause the recipe to fail. The best recipes guide the chef not just in the “model” way of making the dish but also by providing allowable substitutions.

Adapting this model to co-design, a recipe-based curriculum would make clear which elements librarians should carefully follow and retain and which elements can be changed without compromising the goal or outcome of the session. For example, the core spirit of co-design might be described as involving all parties as active participants, with librarians modeling a facilitation role. Some common substitutions might include the specific tools (physical or digital) used for communication and capturing and refining of ideas, and the nature and form of the final deliverable from the co-design experience. Framing co-design in this way also allows the co-design team to ask a more fundamental question: What is most important for the specific project, the process of cooking together or the dish produced? In other words, is the project more focused on the collaborative process itself, the design outcome, or both? If the process is more important, we can substitute even more to produce a different (but equally fine) dish than originally anticipated. In other cases, producing a particular dish may be important and the cook needs to ensure the main ingredients remain intact. In the context of co-design, for instance, is it okay if the team ends up producing a collaborative art creation activity instead of a VR game or not? Thinking about co-design with the recipe-based model can help the librarians balance project structure and flexibility in a more practical manner. We expect the co-design recipe to specify the following elements:

- Dish: the design goal and outcome
- Ingredients: elements needed for preparing the co-design session
 - Main ingredients: things that cannot be substituted, such as the core spirit of co-design or specific participants
 - Substitutes: things that can be replaced, such as alternative co-design approaches, strategies, and activities. Ideally, a number of alternatives should be provided.
 - Local ingredients: strengths of the specific libraries or librarian leading the co-design, such as expertise/knowledge in specific domain, strong rapport with youth participants, etc.
- Cooking tools: tools and technologies needed for the co-design session
 - Required tools: critical tools/technologies that must be included
 - Optional tools: tools and technologies that can be replaced. Ideally, it should provide a number of optional tools/technologies.

An example of a recipe-based co-design may look like table 1.

Table 1. Recipe-Based Co-design with a Food Analogy of Mac and Cheese

Dish (goal/outcome)		Mac and cheese	Envisioning a VR space for supporting youth mental health
Ingredients	Main ingredients	Cheese macaroni	Core spirit of co-design (partnership, inquiry-based learning); youth participants; design sprints
	Substitutes	Types of macaroni Types of cheese	Affinity diagram; user journey map; storyboard; mind map
	Local ingredients	Bacon	Collaboration with a local community support group for youth mental health; involving interns from local universities
		Kimchi Curry Pulled pork	
Cooking tools	Required tools	Pot of water	VR gear, computer, physical space (for in-person); internet connectivity (for online)
	Optional tools	Colander/strainer Spoon/spatula	VR program (e.g., Liminal, Happy Place, TRIPP); stickies; paper/pen OR digital canvas (e.g., MIRO, Mural)

Conclusion

Through VR co-design sessions involving teens at multiple public libraries, we were able to identify opportunities and challenges related to designing and implementing such programs, especially related to supporting teen mental health. While mental health was initially perceived by librarians as an intimidating topic to engage in with teens, the librarians' perceptions shifted throughout the program. Librarians reflected on the roles of libraries and librarians in supporting teen mental health and ultimately showed greater confidence and interest in offering library programming related to mental health and well-being. VR improved the appeal of the program to some teens, and the social aspect of VR was seen as a major draw. Some challenges that arose included identifying appropriate collaboration tools, providing technical support exclusively through online/virtual interaction, and the potential for excluding certain participants. This resulted in mixed responses regarding future VR-related library programming. Unlike VR, co-design was unanimously seen as a valuable approach to allowing teens to actively participate in a project and see their input in the design process reflected in the outcome. However, librarians who lacked prior co-design experience expressed seemingly conflicting desires; they wanted detailed structure to get the project started but were impeded by the level of structure as they encountered unexpected situations.

On the basis of the findings from this study, we propose a recipe-based co-design model as an approach to better support co-design, especially for librarians who are interested in adopting

co-design in their programs but are less familiar with co-design methods. This approach enables librarians to see what core elements need to be preserved in the co-design process and highlights areas where creative adaptation can be deployed as needed to overcome technical and logistical challenges. This may be especially valuable for projects with multiple sites to help ensure consistent quality of programming across the sites, regardless of local contexts and resources.

In our future work, we plan to continue building upon this project and work with other library sites to offer additional co-design programs focused on teens' mental health and well-being. We will be adopting and testing the recipe-based co-design model to investigate whether and how our model can help librarians feel more confident about running a co-design program and do it productively, even if the librarians are new to co-design.

Appendix **Interview Protocol**

Pre-program Interview Questions

Questions about the librarians

1. Please tell us about your work in the library.
 - 1.1. What are your main tasks in your current position?
 - 1.2. How long have you been in that position?

Questions about teen services

2. What kinds of teen services does your library provide? Can you tell us what has been successful so far?
 - 2.1. Do you provide any virtual services? If so, what?
3. How comfortable do you feel about working with teens?
4. What are your biggest challenges in providing teen services at your library?
5. How has the COVID-19 pandemic affected your work and teen services at your library?

Questions about mental health

6. What do you think are the roles of librarians and libraries in supporting patrons' mental health?
7. How comfortable do you feel about running library programs to support patrons' mental health?
8. What kinds of concerns, if any, do you have about offering programs to support patrons' mental health, especially for youth?

Questions about VR

9. How would you describe your own experiences with VR so far?

10. What VR experiences have you tried (what were they called, and what were they about)? How did you find them?
11. What kind of teen support do you think is possible through VR?
12. What do you see as the largest potential contribution VR can offer to teen library patrons?
13. What do you see as the biggest challenges and limitations to using VR for your teen community?

Questions about co-design

14. Have you ever been part of a co-design experience with teens (or in general)? If so, what was it like?
15. Have you ever had any courses or training in design methods in your MLIS program (or university in general)? Do you see yourself as a designer?
16. How might co-design help support your teen patrons?

General questions

17. What benefits do you see stemming from the VRtality project? (anticipated opportunities)
18. What are you most concerned about? (anticipated challenges)
19. What kinds of support would you like for you to be successful in this project?
 - 19.1. What support do you have from your organization? (e.g., goal alignment, human resource, tech support)
20. Is there anything more you'd like to share related to this project?

Post-program Interview Questions

Questions about mental health

1. How has your view regarding the roles of librarians and libraries in supporting patrons' mental health changed?
2. Given your experience, has your level of comfort changed regarding running library programs to support patrons' mental health?
3. Since actually running the program, have any of your concerns changed regarding offering programs to support patrons' mental health?

Questions about VR

4. After the program, what do you think about the potential for a VR experience to be implemented like this? Do you feel that the focus on VR offered unique opportunities? Why or why not?

Questions about co-design

5. Do you think the co-design aspect of the programming (as opposed to other kinds of programming) contributed to participants' engagement? Why or why not?

General questions

6. Reflecting on what you originally anticipated as opportunities, did the program succeed in bringing these opportunities?
7. Reflecting on what you were originally concerned about, did the program in fact encounter those challenges, and do you feel that you were able to overcome them?
8. Given your original concerns and impressions about the project, what has surprised you based on your experience? (How would they do it differently if they had to do it again?)
9. How was the experience of doing this remotely?
10. What kinds of adjustments or changes did you have to make throughout the process to make the co-design successful? (e.g., in terms of setting including tech [hardware and software], interactions with teens [rapport, affect, motivation, etc.], project logistics [design techniques, tools])
11. Is there anything more you'd like to share related to the project?

Community of Practice Post-Project Discussion Prompts

Project recap questions

- What are some lessons learned through your experience implementing this program?
- What are the biggest likes you experienced while implementing this program?
- What are some design ideas you have for making the program more engaging, impactful, and/or easier to implement?

Co-design iceberg

- Around the top of the iceberg, add everything you expected the project would include and require of you.
- Below the iceberg, add everything that was unexpected or different from what you planned the project would look like and/or require of you.

Designing solutions

- Take one of the responses from the bottom of the iceberg, design a solution to that response below, add sticky note ideas, images, or anything else to help explain your solution.

References

- Agosto, Denise E., Rachel M. Magee, Michael Dickard, and Andrea Forte. 2016. "Teens, Technology, and Libraries: An Uncertain Relationship." *Library Quarterly* 86 (3): 248–69.

- American Academy of Pediatrics. 2021. "AAP-AACAP-CHA Declaration of a National Emergency in Child and Adolescent Mental Health." <https://www.aap.org/en/advocacy/child-and-adolescent-healthy-mental-development/aap-aacap-cha-declaration-of-a-national-emergency-in-child-and-adolescent-mental-health/>.
- Björling, Elin A., Rachael Cicero, Aditya Sankar, and Anand Sekar. 2019. "Thought Disposal: Co-Designing a Virtual Interaction to Reduce Stress in Teens." In *Proceedings of the 18th ACM International Conference on Interaction Design and Children*, 562–67. New York: Association for Computing Machinery.
- Braun, Linda W., Nicole A. Cooke, Denise Lyons, Sara Ryan, and Beth Yoke. 2018. *Transforming Library Services for and with Teens through Continuing Education*. Chicago: YALSA.
- Bulanda, Jeffrey J., Christine Bruhn, Trisha Byro-Johnson, and Melissa Zentmyer. 2014. "Addressing Mental Health Stigma among Young Adolescents: Evaluation of a Youth-Led Approach." *Health & Social Work* 39 (2): 73–80.
- Camino, Linda. 2005. "Pitfalls and Promising Practices of Youth-Adult Partnerships: An Evaluator's Reflections." *Journal of Community Psychology* 33 (1): 75–85.
- Clarke, Victoria., Virginia Braun, and Nikki Hayfield. 2015. "Thematic Analysis." In *Qualitative Psychology: A Practical Guide to Research Methods*, 3rd ed., edited by Jonathan A. Smith, 222–48. London: Sage.
- Cleave, Jayne, and John Geijsman. 2020. "LibraryCraft: How the COVID-19 Pandemic Led to the Growth of the WA Libraries Public Minecraft Server." *Digital Library Perspectives* 36 (4): 377–88.
- Courtney, Darren, Priya Watson, Marco Battaglia, Benoit H. Mulsant, and Peter Szatmari. 2020. "COVID-19 Impacts on Child and Youth Anxiety and Depression: Challenges and Opportunities." *Canadian Journal of Psychiatry* 65 (10): 688–91.
- Cox, Andrew, and Liz Brewster. 2020. "Library Support for Student Mental Health and Well-Being in the UK: Before and during the COVID-19 Pandemic." *The Journal of Academic Librarianship* 46 (6): 102256.
- Dahya, Negin, W. E. King, Kung Jin Lee, and Jin Ha Lee. 2021. "Perceptions and Experiences of Virtual Reality in Public Libraries." *Journal of Documentation* 77 (3): 617–37.
- Dalsgaard, Peter, and Eva Eriksson. 2013. "Large-Scale Participation: A Case Study of a Participatory Approach to Developing a New Public Library." In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 399–408. New York: Association for Computing Machinery.
- Druin, Allison, Benjamin B. Bederson, Anne Rose, and Ann Weeks. 2009. "From New Zealand to Mongolia: Co-Designing and Deploying a Digital Library for the World's Children." *Children, Youth and Environments* 19 (1): 34–57.
- Flores, Araceli, Marsha M. Linehan, S. Rob Todd, and Hunter G. Hoffman. 2018. "The Use of Virtual Reality to Facilitate Mindfulness Skills Training in Dialectical Behavioral Therapy for Spinal Cord Injury: A Case Study." *Frontiers in Psychology* 9: 531.
- Friedman, Victor. J. 2006. "Action Science: Creating Communities of Inquiry in Communities of Practice." *Handbook of Action Research: The Concise Paperback Edition*, edited by Peter Reason and Hilary Bradbury-Huang, 131–43. London: SAGE.
- Frost, Megan, Michael Goates, Sarah Cheng, and Jed Johnston. 2020. "Virtual Reality: A Survey of Use at an Academic Library." *Information Technology and Libraries* 39 (1): 1–12.
- Geiger, A. W. 2017. "Most Americans—Especially Millennials—Say Libraries Can Help Them Find Reliable, Trustworthy Information." <https://www.pewresearch.org/fact-tank/2017/08/30/most-americans-especially-millennials-say-libraries-can-help-them-find-reliable-trustworthy-information/>.
- Giota, Joanna, and Jan-Eric Gustafsson. 2017. "Perceived Demands of Schooling, Stress and Mental Health: Changes from Grade 6 to Grade 9 as a Function of Gender and Cognitive Ability." *Stress and Health* 33 (3): 253–66.

- Goddard, Jon. 2020. "Public Libraries Respond to the COVID-19 Pandemic, Creating a New Service Model." *Information Technology and Libraries* 39 (4): 1–4.
- Golberstein, Ezra, Hefei Wen, and Benjamin F. Miller. 2020. "Coronavirus Disease 2019 (COVID-19) and Mental Health for Children and Adolescents." *JAMA Pediatrics* 174 (9): 819–20.
- Howard, Vivian. 2011. "What Do Young Teens Think About the Public Library?" *Library Quarterly* 81 (3): 321–44.
- Ivey-Stephenson, Asha Z., Zewditu Demissie, Alexander E. Crosby, Deborah M. Stone, Elizabeth Gaylor, Natalie Wilkins, Richard Lowry, and Margaret Brown. 2020. "Suicidal Ideation and Behaviors among High School Students – Youth Risk Behavior Survey, United States, 2019." *Morbidity and Mortality Weekly Report. Supplement* 69 (1): 47–55.
- Kretz, Chris, Claire Payne, and Dana Reijerkerk. 2021. "Study Room Time Machine: Creating a Virtual Library Escape Game during COVID." *College & Undergraduate Libraries* 28 (3–4): 273–95.
- Lee, Kung Jin, W. E. King, Negin Dahya, and Jin Ha Lee. 2020. "Librarian Perspectives on the Role of Virtual Reality in Public Libraries." *Proceedings of the Association for Information Science and Technology* 57 (1): e254.
- Lee, Kung Jin, Wendy Roldan, Tian Qi Zhu, Harkiran Kaur Saluja, Sungmin Na, Britnie Chin, Yilin Zeng, Jin Ha Lee, and Jason Yip. 2021. "The Show Must Go On: A Conceptual Model of Conducting Synchronous Participatory Design with Children Online." In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. New York: Association for Computing Machinery.
- Lessick, Susan, and Michelle Kraft. 2017. "Facing Reality: The Growth of Virtual Reality and Health Sciences Libraries." *Journal of the Medical Library Association* 105 (4): 407–8.
- Maloney, Divine, Guo Freeman, and Andrew Robb. 2021. "Stay Connected in An Immersive World: Why Teenagers Engage in Social Virtual Reality." In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference*, 69–79. New York: Association for Computing Machinery.
- National Institute of Mental Health (NIMH). 2019. "Major Depression." <https://www.nimh.nih.gov/health/statistics/major-depression>.
- Pope, Hannah. 2018. "Chapter 2: Incorporating Virtual and Augmented Reality in Libraries." *Library Technology Reports* 54 (6): 8–11.
- Power, E., S. Hughes, D. Cotter, and M. Cannon. 2020. "Youth Mental Health in the Time of COVID-19." *Irish Journal of Psychological Medicine* 37 (4): 301–5.
- Racine, Nicole, Brae Anne McArthur, Jessica E. Cooke, Rachel Eirich, Jenney Zhu, and Sheri Madigan. 2021. "Global Prevalence of Depressive and Anxiety Symptoms in Children and Adolescents during COVID-19: A Meta-Analysis." *Archives of Pediatrics & Adolescent Medicine* 175 (11): 1142–50.
- Sanders, Elizabeth B.-N. 2002. "From user-Centered to Participatory Design Approaches." In *Design and the Social Sciences*, edited by Jorge Frascara, 18–25. London: CRC Press.
- Santos Green, Lucy, Stephanie A. Jones, and Andrea Burke Panne. 2017. "School Librarians Fully Online: Preparing the Twenty-First Century Professional." https://www.ala.org/aasl/sites/ala.org.aasl/files/content/aaslpubsandjournals/slr/vol20/SLR_SchoolLibFullyOnline_V20.pdf.
- Schön, Donald A. 1987. *Educating the Reflective Practitioner: Toward a New Design for Teaching and Learning in the Professions*. San Francisco: Jossey-Bass.
- Senge, Peter M. 1991. "Team Learning." *The McKinsey Quarterly* 2: 82–94.
- Somerville, Mary M., and Lydia Collins. 2008. "Collaborative Design: A Learner-Centered Library Planning Approach." *Electronic Library* 26 (6): 803–20.
- Somerville, Mary M., and Zaana Howard. 2010. "'Information in Context': Co-Designing Workplace Structures and Systems for Organisational Learning." *Information Research* 15 (4).

- Subramaniam, Mega. 2016. "Designing the Library of the Future for and with Teens: Librarians as the 'Connector' in Connected Learning." *Journal of Research on Libraries and Young Adults* 7 (2): 1–18.
- US Department of Health and Human Services. 2001. *Mental Health: Culture, Race, and Ethnicity – A Supplement to Mental Health: A Report of the Surgeon General*. Rockville, MD: US Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Center for Mental Health Services.
- Wang, Jingyi, Hao Wang, Haijiang Lin, Marcus Richards, Shuyue Yang, Hongbiao Liang, Xiaoxiao Chen, and Chaowei Fu. 2021. "Study Problems and Depressive Symptoms in Adolescents during the COVID-19 Outbreak: Poor Parent-Child Relationship as a Vulnerability." *Globalization and Health* 17 (1): 40.
- Yip, Jason C., Kung Jin Lee, and Jin Ha Lee. 2020. "Design Partnerships for Participatory Librarianship: A Conceptual Model for Understanding Librarians Co Designing with Digital Youth." *Journal of the Association for Information Science and Technology* 71 (10): 1242–56.
- Yip, Jason, and Kung Jin Lee. 2018. "The Design of Digital Learning Activities for Libraries through Participatory Design." In *Reconceptualizing Libraries*, edited by Victor R. Lee and Abigail L. Phillips, 203–16. New York: Routledge.

Jin Ha Lee: professor and founding director of the GAMER (GAME Research) Group at the University of Washington Information School. Lee's research focuses on exploring new ideas and approaches for organizing and providing access to popular music, multimedia, and interactive media, understanding user behavior related to the creation and consumption of these media, and using these media for informal learning in venues such as libraries and museums. Email: jinhalee@uw.edu.

Elin Björling: senior research scientist in Human Centered Design and Engineering at the University of Washington. Björling utilizes participatory design to explore new technologies to support mental health and education for youth and children. Her recent research has focused primarily on the co-design and development of social robots. As the project lead for Project EMAR, Björling is leading an interdisciplinary team to build a social robot to capture stress and moods from teens. She is also a trained evaluator, having developed mixed-methods study designs for large federal projects in education and health. Email: bjorling@uw.edu.

Ryan McLean: recent masters graduate of the Human Centered Design and Engineering program at the University of Washington. McLean completed his graduate studies while simultaneously working as a manufacturing engineer in the aerospace sector. In addition to virtual reality, he has applied human-centered principles to research and design problems, including voice user interfaces and sustainable nonprofit funding. He is particularly interested in how these technologies and others, when adopted in a human-centric way, can improve outcomes in fields such as health care, manufacturing, and education. Email: rsmclean@uw.edu.

Anna Moreno: recent masters graduate from the Human Centered Design and Engineering program at the University of Washington. Moreno currently works as a UX Researcher at ServiceNow. Her research interests include health technology, responsible AI, and participatory design. Prior to graduate school, she worked as a brand strategist within the advertising industry, where she gained

exposure to consumer behavior research and the profound impact of social media platforms. Email: moreanna@gmail.com.

Kung Jin Lee: assistant professor, department of Library and Information Science at Ewha Womans University. Prior to joining the faculty at Ewha, Lee worked on projects as a researcher on understanding VR in libraries, conducting participatory design in libraries and online. Her research focuses primarily on utilizing participatory design methods with multiple stakeholders for curriculum development and technology design. Her work has been published in *ACM CHI*, *Journal of the Association for Information Science and Technology*, and *Journal of Documentation*. She received her PhD in Information Science from the University of Washington. Email: kungjinlee@ewha.ac.kr.